

DAY 29 – INTERNSHIP JOURNEY REFLECTION

Personal Learning Roadmap (90 Days)

Date: 09 October 2026

1. Task

1) Write a reflection summarizing the complete internship journey, including technical skills acquired, major challenges overcome, key achievements, lessons learned, and a personal learning roadmap for the next three months.

2) Personal Learning Roadmap (90 Days)

2. Work Done

Part 1: Internship Journey Reflection

Prepared a reflection summarizing my AI/ML internship journey, including technical skills acquired, challenges faced, achievements, and lessons learned.

Technical Skills Acquired

- Python programming and data structures.
- Data analysis using NumPy and Pandas.
- Data visualization using Matplotlib and Seaborn.
- Machine learning fundamentals using Scikit-learn, including Logistic Regression and Decision Trees.
- Basic understanding of LLMs, Transformers, and Retrieval-Augmented Generation (RAG).
- AI agents, multi-agent workflows, function calling, and LangGraph concepts.
- FastAPI and asynchronous API development.
- PostgreSQL, database replication concepts, and SQLAlchemy.
- Docker, Git, API testing, and backend development concepts.
- Authentication, Role-Based Access Control (RBAC), and secure backend operations.

Major Challenges Overcome

- Resolving Python library and dependency compatibility issues.
- Understanding asynchronous APIs and comparing synchronous and asynchronous performance.
- Troubleshooting database configuration, PostgreSQL replication, and migration issues.
- Understanding how AI agents communicate, select tools, and execute workflows.
- Designing secure chatbot operations using authentication and permission validation.
- Organizing project documentation and explaining technical architecture clearly.

Key Achievements

- Completed practical exercises involving Python, data analysis, and machine learning.
- Explored basic RAG pipelines and LLM-based applications.
- Practiced function calling and multi-agent workflow design.
- Converted a synchronous API into an asynchronous API and benchmarked the results.
- Worked with PostgreSQL configuration and database replication concepts.
- Worked on the Linkific HR Automation Platform.
- Selected and reviewed the Role-Based Chatbot Task Assignment feature.

- Prepared project presentation slides, README documentation, architecture documentation, and technical interview Q&A.;

Lessons Learned

- Practical implementation is important for understanding technical concepts.
- Debugging is an essential part of software development.
- Modular code improves readability, testing, and maintenance.
- AI applications need proper validation and secure backend controls.
- Authentication and authorization are essential for applications with different user roles.
- Clear documentation and communication help explain technical work.

Part 2: Personal Learning Roadmap (90 Days)

Prepared a three-month roadmap focused on Python, backend development, AI/ML, AI agents, database management, testing, and deployment. It prioritizes strong fundamentals, practical implementation, and a complete project.

3. Challenges Faced

- 1. Reviewing the complete journey:** Organizing different topics and practical exercises into a structured reflection.
- 2. Identifying areas for improvement:** Identifying skills needing more practice, especially backend development, AI agent workflows, database management, and security.
- 3. Planning the next 90 days:** Prioritizing topics into manageable phases while balancing theory, coding, and project work.
- 4. Connecting learning with practice:** Finding ways to apply internship concepts to real-world projects instead of relying only on theory.

4. Outcome

Prepared a structured internship reflection and identified technical skills, achievements, challenges, and lessons learned. Established a 90-day learning roadmap to improve technical knowledge, practical implementation, problem-solving, and project development.

PERSONAL LEARNING ROADMAP (90 DAYS)

Duration: 9 October 2026 – 6 January 2027

Month 1: Strengthen Technical Fundamentals (Days 1–30)

Focus: Python, SQL, backend development, and software engineering fundamentals.

Topics to Learn

- Python fundamentals, functions, OOP, and exception handling.
- Data structures and basic problem-solving.
- SQL queries, joins, indexes, and database relationships.
- FastAPI, REST APIs, request validation, and error handling.
- SQLAlchemy and database integration.
- Git, GitHub, and clean coding practices.

Practical Activities

- Build REST APIs using FastAPI.
- Connect FastAPI to PostgreSQL.
- Implement CRUD operations.
- Write basic unit tests for API endpoints.
- Practice SQL problems regularly.

Expected Outcome: Develop a stronger foundation in Python, SQL, and backend development.

Month 2: AI/ML and AI Agent Development (Days 31–60)

Focus: Machine learning, LLMs, RAG, and AI agents.

Topics to Learn

- Data preprocessing and model evaluation.
- Classification algorithms and performance metrics.
- LLM fundamentals and prompt engineering.
- Embeddings, vector databases, and RAG.
- Function calling and tool integration.
- LangChain and LangGraph fundamentals.
- Multi-agent workflows and state management.

Practical Activities

- Build a basic machine learning application.
- Develop a document-based RAG application.
- Create an AI agent that uses tools.
- Build a multi-step workflow with clear responsibilities for each agent.
- Add error handling and basic tests to AI workflows.

Expected Outcome: Gain practical experience building AI-powered applications and agent-based workflows.

Month 3: Project Development and Production Readiness (Days 61–90)

Focus: Building, testing, securing, and deploying a complete application.

Topics to Learn

- Backend architecture and modular code organization.
- Authentication and Role-Based Access Control.
- PostgreSQL optimization and database management.
- Unit testing and API testing.
- Docker and application deployment.
- Logging, monitoring, and error handling.
- API security and environment variable management.
- Technical documentation and interview preparation.

Practical Activities

- Build or improve a complete AI-powered application.
- Integrate an AI chatbot with backend tools.
- Implement authentication and permission checks.
- Connect the application to PostgreSQL.
- Write unit and API tests.
- Containerize the application using Docker.
- Prepare a README, architecture diagram, and project demonstration.
- Practice technical interview questions related to the project.

Expected Outcome: Complete a practical AI-powered project demonstrating backend development, AI integration, database management, security, testing, and deployment.

5. Final Goal

- Strengthen Python and backend development skills.
- Build practical AI/ML and AI agent applications.
- Improve database design and API security knowledge.
- Become more confident in debugging and problem-solving.
- Complete a well-documented, deployable project.
- Improve readiness for software engineering and AI/ML-related job opportunities.

By the end of the 90-day roadmap, the goal is to turn internship learning into practical, demonstrable skills through consistent practice and one complete project.